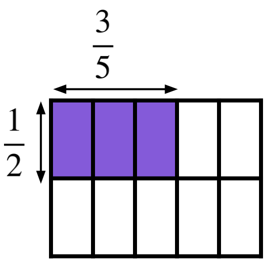
Checking understanding of multiplication with fractions

Task A: Exploring the area model

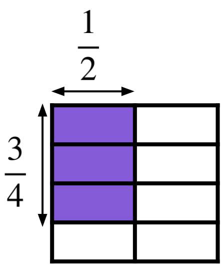
1) Each of the large squares have an area of 1 m^2. For each square:

a) Label the dimensions of the shaded rectangle
b) Write the calculation represented by the area of the shaded rectangle.
c) Work out the area of the square that is shaded.



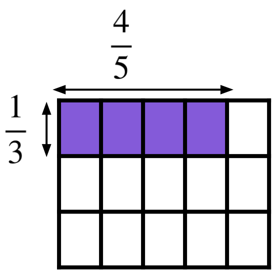
$\frac{1}{2} \times \frac{3}{5} = \frac{3}{10}$

$\frac{3}{10}$ m^2



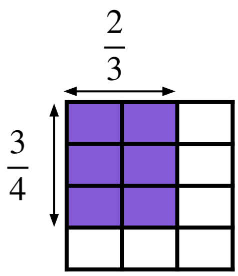
$\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$

$\frac{3}{8}$ m^2



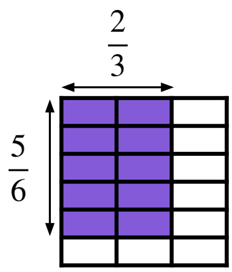
$\frac{1}{3} \times \frac{4}{5} = \frac{4}{15}$

$\frac{4}{15}$ m^2



$\frac{3}{4} \times \frac{2}{3} = \frac{6}{12}$

$\frac{6}{12}$ m^2 $= \frac{1}{2}$ m^2



$\frac{5}{6} \times \frac{2}{3} = \frac{10}{18}$

$\frac{10}{18}$ m^2 $= \frac{5}{9}$ m^2

Task B: Multiplication of fractions

1) Calculate the answers to the following. Simplify all answers. Find them in the table and write down which letter each gives. Rearrange your letters to make a word.

A	B	C	D	E	F	G	H	I	J	K	L	M
$\frac{1}{4}$	$\frac{19}{20}$	$\frac{4}{5}$	$\frac{7}{8}$	$\frac{7}{10}$	$\frac{7}{9}$	$\frac{1}{8}$	$\frac{1}{5}$	$\frac{13}{20}$	$\frac{3}{8}$	$\frac{1}{12}$	$\frac{1}{10}$	$\frac{3}{4}$
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$\frac{1}{2}$	$\frac{5}{8}$	$\frac{5}{12}$	$\frac{4}{7}$	$\frac{1}{3}$	$\frac{7}{12}$	$\frac{3}{5}$	$\frac{2}{5}$	$\frac{2}{9}$	$\frac{3}{11}$	$\frac{3}{10}$	$\frac{1}{6}$	$\frac{5}{6}$

a) $\frac{1}{4}$ A

b) $\frac{1}{3}$ R

c) $\frac{2}{5}$ U

d) $\frac{3}{4}$ M

e) $\frac{5}{8}$ O

f) $\frac{1}{2}$ N

g) $\frac{1}{3}$ R

h) $\frac{7}{10}$ E

i) $\frac{3}{5}$ T

The word is NUMERATOR

2) What are the missing digits in the following calculations?

a) $\frac{2}{3} \times \frac{4}{7} = \frac{\boxed{8}}{21}$

b) $\frac{3}{10} \times \frac{3}{5} = \frac{9}{\boxed{50}}$

c) $\frac{\boxed{3}}{10} \times \frac{7}{8} = \frac{21}{80}$

d) $\frac{3}{5} \times \frac{2}{\boxed{5}} = \frac{6}{25}$

e) $\frac{2}{3} \times \frac{5}{8} = \frac{5}{\boxed{12}}$

f) $\frac{3}{5} \times \frac{4}{\boxed{9}} = \frac{4}{15}$

g) $\frac{6}{11} \times \frac{\boxed{5}}{9} = \frac{10}{33}$

h) $\frac{\boxed{7}}{12} \times \frac{3}{5} = \frac{7}{20}$